ALGEBRA 1 · UNIT 5: LINEAR FUNCTIONS AND GRAPHING

Mountain Trail Designer

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students act as trail designers for a ski resort. They calculate slopes from a trail profile map, rate trails as green, blue, or black, write equations for trails and the chairlift, and design a new route so beginners can reach the base safely.

Slope becomes a design constraint. Students compute slopes from coordinates, compare steepness using absolute value, and discover that no all-beginner route exists. They then design a new route, which requires working backwards from slope requirements.

Standards Addressed

- F-LE.1b Recognize situations with a constant rate of change
- F-IF.6 Calculate and interpret the average rate of change
- A-CED.2 Create equations in two variables
- G-GPE.5 Use slopes of parallel and perpendicular lines (CP1 and Honors)

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch: show a real trail map and ask what makes a trail 'black diamond'. Part 1 slopes and ratings.
Day 2	Part 2 equations of trails and the chairlift, plus rate of change from a table.
Day 3	Part 3 design a new trail and create the map poster.

Materials

- Student handout with the trail map
- Calculators
- Colored pencils (green, blue, black)
- Graph paper or poster paper for the new trail map

How the Three Versions Differ

Level	What changes
CP2	Guided slope table with rise and run columns, rating card, slope-intercept equations with a starter, and an open design with one new point.
CP1	All trails rated without scaffolding, point-slope equations, parallel and perpendicular lines, and a design that must meet the rating rules.
Honors	Ratings written as absolute value inequalities, percent grade, standard form, and a design where the new point's elevation must satisfy a compound inequality.

Watch For (Common Misconceptions)

- Computing run ÷ rise instead of rise ÷ run.
- Thinking a more negative slope is 'less steep'. Use steepness (absolute value).
- Using the summit's y-value as the y-intercept for a trail that doesn't start at x = 0.

Extension Ideas

- Find a real ski resort's trail data and compute slopes.
- Convert slope to angle with a protractor drawing.
- Add a snowboard terrain park with a slope inequality.

Answer Key – CP2

Part 1: Rate the Trails

#	Task	Answer
1	Use the rating card to answer: What rating does a trail with a slope of −0.3 get? Explain how you decided.	Blue, because 0.3 is more than 0.25 and at most 0.4.
2	Complete the table. The first row is done for you.	Plunge: rise −12, run 20, slope −0.6, Black. Easy Street: −10, 40, −0.25, Green. Meadow: −4, 40, −0.1, Green. Finale: −12, 20, −0.6, Black. Long Haul: −16, 60, about −0.27, Blue.
3	Which trail is the steepest? Which is the least steep? Use the slopes to explain.	Steepest: The Plunge and Finale (steepness 0.6). Least steep: Meadow (0.1).
4	A beginner wants to ski from the Summit to the Base Lodge using only green and blue trails. Is it possible? Use the map to explain.	No. Every route to the Base Lodge uses at least one black trail (The Plunge, Cliffside, or Finale). The only blue trail into the base (Long Haul) starts at Midway, and you can only reach Midway on The Plunge.

Part 2: Trail Equations

#	Task	Answer
5	Ridge Run starts at (0, 48). Write its equation. What does the 48 mean on the mountain?	$y = -0.4x + 48$. The 48 means the summit is 4,800 feet high.
6	The chairlift runs straight from the Base Lodge (100, 12) to the Summit (0, 48). Find its slope and write its equation.	Slope = $(48 - 12) \div (0 - 100) = 36 \div (-100) = -0.36$; $y = -0.36x + 48$
7	Use your chairlift equation to find the elevation of the lift at x = 50.	$y = -0.36(50) + 48 = 30$, so 3,000 feet.
8	A snow groomer climbs the mountain. Find its rate of change and write an equation for its elevation y (in feet) after t minutes. When will it…	Rate = $300 \div 5 = 60$ ft per minute; $y = 60t + 1200$. $60t + 1200 = 4800 \rightarrow t = 60$ minutes.

Part 3: Design a New Trail

#	Task	Answer
9	Add one new point to the map (your choice). Draw a new trail from an existing point to your new point, and a trail from your new point to the Base…	Answers vary. Example: new point (120, 12) as a second base: Deer Hollow (80, 24) to (120, 12).
10	Find the slope and rating of each new trail. Both must be green or blue.	Example: $(12 - 24) \div (120 - 80) = -0.3 \rightarrow$ Blue.

11	Give your trails names and write the full beginner route from the Summit to the base.	Example: Ridge Run → Easy Street → Switchback → new trail.

Scoring note: In the profile map, a trail may go to a new base point to the right of x = 100. Accept any design whose slopes are correctly computed and rated.

Answer Key – CP1

Part 1: Rate the Trails

#	Task	Answer
1	Find the slope, steepness, and rating of every trail.	Ridge Run −0.4, 0.4, Blue; Plunge −0.6, 0.6, Black; Long Haul ≈ −0.27, 0.27, Blue; Easy Street −0.25, 0.25, Green; Cliffside −0.45, 0.45, Black; Meadow −0.1, 0.1, Green; Switchback −0.3, 0.3, Blue; Finale −0.6, 0.6, Black.
2	Explain why a slope of −0.6 is steeper than a slope of −0.25 even though −0.6 is less than −0.25.	Steepness is the size of the slope (absolute value). $\|-0.6\| = 0.6 > 0.25$, so the trail drops more per unit of horizontal distance.
3	Is there a route from the Summit to the Base Lodge with no black trails? Justify with your table.	No. Every route into the Base Lodge uses Finale, Cliffside, or Long Haul, and Long Haul can only be reached from Midway by The Plunge (black).
4	Ridge Run drops 800 feet. How many feet does a skier travel horizontally on Ridge Run? Use the slope to explain.	Drop 8 hundred feet; 8 ÷ 0.4 = 20 hundred feet = 2,000 feet horizontally.

Part 2: Trail & Lift Equations

#	Task	Answer
5	Write the equation of Easy Street using point-slope form, then rewrite it in slope-intercept form.	$y - 40 = -0.25(x - 20) \rightarrow y = -0.25x + 45$
6	Write the equation of the chairlift from (100, 12) to (0, 48). At what x-value is the lift at an elevation of 2,400 feet?	$y = -0.36x + 48$. $24 = -0.36x + 48 \rightarrow x = 66.67$ (about 6,667 ft from the summit).
7	A new snow fence must run perpendicular to Ridge Run and pass through the Ridge (20, 40). Write its equation.	Perpendicular slope = 2.5; $y - 40 = 2.5(x - 20) \rightarrow y = 2.5x - 10$
8	A snow groomer's elevation is shown in the table. Write an equation and find when it reaches 4,800 feet.	Rate = 240 ÷ 4 = 60 ft per minute; $y = 60t + 1200$. $60t + 1200 = 4800 \rightarrow t = 60$ minutes.

Part 3: Design a Beginner Route

#	Task	Answer
9	Add one or two new points. Draw your new trails on the map and give the coordinates.	Answers vary.
10	Show the slope and rating of every trail on your route.	Answers vary.

11	Write the equation of each new trail. One new trail must be parallel to an existing trail. Prove it with slopes.	Answers vary; parallel trails have equal slopes.

Answer Key – Honors

Part 1: Ratings as Inequalities

#	Task	Answer
1	Write each rating as an inequality using $\|m\|$.	Green: $\|m\| \le 0.25$. Blue: $0.25 < \|m\| \le 0.4$. Black: $\|m\| > 0.4$.
2	Find the slope and rating of every trail. Record them in the table.	Ridge Run −0.4 Blue; Plunge −0.6 Black; Long Haul ≈ −0.267 Blue; Easy Street −0.25 Green; Cliffside −0.45 Black; Meadow −0.1 Green; Switchback −0.3 Blue; Finale −0.6 Black.
3	Ski resorts often give the grade as a percent: grade = $100\|m\|$%. Find the grade of the steepest and least steep trails.	Steepest: 60% (Plunge, Finale). Least steep: 10% (Meadow).
4	Prove that no all-green-or-blue route exists from the Summit to the Base Lodge.	Every trail ending at the Base Lodge is Long Haul, Cliffside, or Finale. Cliffside and Finale are black. Long Haul starts at Midway, whose only incoming trail (The Plunge) is black. So every route uses a black trail.

Part 2: Equations & Relationships

#	Task	Answer
5	Write the equation of the chairlift in slope-intercept form and in standard form $Ax + By = C$ (A, B, and C integers).	$y = -0.36x + 48$; multiply by 25: $25y = -9x + 1200 \rightarrow 9x + 25y = 1200$
6	Show that the Ridge Run line and the Midway-to-Deer Hollow line are not parallel, and find where the two lines (extended) intersect.	Slopes −0.4 and −0.1 are different, so not parallel. Ridge Run: $y = -0.4x + 48$. Meadow: $y = -0.1x + 32$. $-0.4x + 48 = -0.1x + 32 \rightarrow x = 53.33$, $y = 26.67$.
7	A snow fence through the Cabin (60, 30) must be perpendicular to Cliffside. Write its equation in standard form.	Cliffside slope −0.45 → perpendicular slope $20 \div 9$ (≈ 2.22). $y - 30 = (20 \div 9)(x - 60) \rightarrow 9y - 270 = 20x - 1200 \rightarrow 20x - 9y = 930$
8	The groomer's elevation is a linear function of time with $f(2) = 1{,}320$ and $f(14) = 2{,}040$. Write $f(t)$ and find the time when $f(t) = 4{,}800$.	Slope $= 720 \div 12 = 60$; $f(t) = 60t + 1200$. $60t + 1200 = 4800 \rightarrow t = 60$ minutes.

Part 3: Engineer an All-Blue Route

#	Task	Answer
9	Let E = (60, y). Write a compound inequality for the steepness of Ridge-to-E so that it is blue, and solve for y.	Steepness $= (40 - y) \div 40$. $0.25 < (40 - y) \div 40 \le 0.4 \rightarrow 10 < 40 - y \le 16 \rightarrow 24 \le y < 30$
10	Do the same for E-to-Base Lodge.	Steepness $= (y - 12) \div 40$. $0.25 < (y - 12) \div 40 \le 0.4 \rightarrow 10 < y - 12 \le 16 \rightarrow 22 < y \le 28$

11	Find all values of y that make both trails blue. Choose one value, write both trail equations, and add the route to your map with a recommendation.	Both: $24 \le y \le 28$. Example y = 26: Ridge to E: $y = -0.35x + 47$; E to Base: $y = -0.35x + 47$ (same line when y = 26!). Other values give two different lines.

Scoring note: Interesting moment: E = (60, 26) lies on the straight line from the Ridge to the Base Lodge (slope −0.35, blue). Ask students why the direct line also works.

Scoring Rubric (all levels)

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	New trail meets the rating requirements, with slopes and equations shown. Goes beyond the minimum with a creative, realistic choice.	New trail meets the rating requirements, with slopes and equations shown.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Trail map is labeled with points, slopes, ratings, and equations. Polished, organized, and visually engaging.	Trail map is labeled with points, slopes, ratings, and equations. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.